

```
In [1]: x=int(input("enter a number"))
        y=int(input("enter second number"))
        z=x+y
        print("sum is ",z)
```

```
enter a number3
enter second number5
sum is 8
```

```
In [2]: x=float(input("enter a number"))
        y=float(input("enter second number"))
        z=x+y
        print("sum is ",z)
```

```
enter a number4
enter second number5
sum is 9.0
```

```
In [3]: x=str(input("enter a number  "))
        y=str(input("enter second number  "))
        z=x+y
        print("sum is ",z)
```

```
enter a number 5
enter second number 4
sum is 54
```

```
In [8]: x=eval(input("enter a number  "))
        y=eval(input("enter second number  "))
        z=x+y #addition
        w=x-y #subtraction
        m=x*y #multiplication
        d=x/y #division
        print("sum is ",z)
        print("difference",w)
        print("product is",m)
        print("division is",d)
```

```
enter a number 10
enter second number 2
sum is 12
difference 8
product is 20
division is 5.0
```

```
In [5]: x=10
        y=2
        z=x+y
        print(z)
```

```
In [6]: type(10)#datatypes
```

```
Out[6]: int
```

```
In [10]: # modulus operator, reminder  
10%3
```

```
Out[10]: 1
```

```
In [11]: 25%3
```

```
Out[11]: 1
```

```
In [12]: 25%5
```

```
Out[12]: 0
```

```
In [14]: 42%2 #even numbers are divisible by 2, reminer 0
```

```
Out[14]: 0
```

```
In [15]: 43%2 # odd not divi, remin 1
```

```
Out[15]: 1
```

```
In [16]: 10//3 #quotient
```

```
Out[16]: 3
```

```
In [17]: 10/3
```

```
Out[17]: 3.3333333333333335
```

```
In [18]: 10%3
```

```
Out[18]: 1
```

```
In [19]: 10//3 #floor division
```

```
Out[19]: 3
```

```
In [20]: 5*5
```

```
Out[20]: 25
```

```
In [21]: 5**2 #exponent operator
```

```
Out[21]: 25
```

```
In [22]: 2**4
```

```
Out[22]: 16
```

```
In [23]: 5**3
```

```
Out[23]: 125
```

```
In [24]: #comparision operator
```

```
In [25]: 3<10
```

```
Out[25]: True
```

```
In [26]: 3>10
```

```
Out[26]: False
```

```
In [28]: 7<=7
```

```
Out[28]: True
```

```
In [30]: 15>=15
```

```
Out[30]: True
```

```
In [31]: 10==10
```

```
Out[31]: True
```

```
In [32]: 4==5 #equal
```

```
Out[32]: False
```

```
In [33]: 10!=10 #not eqaul
```

```
Out[33]: False
```

```
In [34]: (10<3) and (2<5)  
# false * true  
# 0 * 1  
#0
```

```
Out[34]: False
```

```
In [35]: (2>5) and (10>34)
```

```
Out[35]: False
```

In []:

```
In [36]: (10<3) or (2<5)
# false + true
# 0 + 1
#1
```

Out[36]: True

```
In [38]: (2<5) or (10<34)
```

Out[38]: True

```
In [40]: not(2<5) #not(true) --- false
```

Out[40]: False

```
In [41]: #membership in , not in
```

```
In [47]: "p" in "python"
```

Out[47]: True

```
In [48]: "p" not in "python"
```

Out[48]: False

```
In [43]: "P" in "python"
```

Out[43]: False

```
In [44]: "py" in "python"
```

Out[44]: True

```
In [45]: "pn" in "python"
```

Out[45]: False

```
In [46]: "n" in "python"
```

Out[46]: True

```
In [49]: # identity operator is, is not
```

```
In [50]: 7==7
```

Out[50]: True

```
In [51]: 7 is 7
```

```
<>:1: SyntaxWarning: "is" with a literal. Did you mean "=="?  
<>:1: SyntaxWarning: "is" with a literal. Did you mean "=="?  
C:\Users\HP\AppData\Local\Temp\ipykernel_6740\3255664600.py:1: SyntaxWarning:  
"is" with a literal. Did you mean "=="?  
7 is 7
```

```
Out[51]: True
```

```
In [53]: "apple" is not "Apple"
```

```
<>:1: SyntaxWarning: "is not" with a literal. Did you mean "!="?  
<>:1: SyntaxWarning: "is not" with a literal. Did you mean "!="?  
C:\Users\HP\AppData\Local\Temp\ipykernel_6740\3624040571.py:1: SyntaxWarning:  
"is not" with a literal. Did you mean "!="?  
"apple" is not "Apple"
```

```
Out[53]: True
```

```
In [57]: #"a" =97  
        #"A" =65 ascii values
```

```
In [56]: "a" > "A"
```

```
Out[56]: True
```

```
In [58]: "AP"<"Ap"
```

```
Out[58]: True
```

```
In [61]: #escape sequence  
        print("sunday's class")
```

```
sunday's class
```

```
In [64]: print("she said 'hello'")
```

```
she said 'hello'
```

```
In [66]: print("she said \"hai\" ") # \" quotes
```

```
she said "hai"
```

```
In [67]: print("she said \"hello ")
```

```
she said "hello
```

```
In [75]: print("she said \"hai\" \"hello\"")
```

```
she said "hai" "hello"
```

```
In [73]: print("hello!")
```

```
hello!
```

```
In [78]: print("firstline \n seconline") #newline
```

```
firstline
seconline
```

```
In [79]: print("firstline \t seconline") #tab
```

```
firstline      seconline
```

```
In [ ]:
```